

Causal Doob packets, payload-comparable resonance, and the coefficient-dominant boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-076 left three proposed largest-root classes: a fresh Gaussian root, a largest explicit control payload, and a coefficient-only balanced root. The proposal correctly identified the causal issue, but it was not yet a disjoint theorem. Ties between coefficient and payload inputs, and Wick contractions which remove a nominal top Gaussian pair, make a raw-monomial ownership rule unsafe.

This note replaces that proposal by two exact operations performed in a fixed order.

First, every complete terminal control increment is decomposed into its strict-past conditional mean and complete Doob differences. The latter are the canonical fresh-Gaussian packets. Their signed expectation is exactly zero at every cutoff. The cancellation includes the current square, coefficient Wick term, both restored first variations, every R-063 lower chaos, and the trace companion; none of those terms centers separately.

Second, inside the remaining predictable endpoint algebra, the only new R-076 shifted resonance is split by smooth Littlewood–Paley input indices. If the multiplier/current resonance scale is no larger than the largest explicit payload scale, the complete block is bounded by the sharpened R-076 Besov theorem and the already available fifteenth moment. The complement is one genuine coefficient-dominant high–high-to-low form.

Thus the old three proposed estimates are replaced by one exact cancellation, one proved absolute estimate, and one smaller signed successor. The last successor, controlled-shell one-use, the $q = 10/9$ Nelson theorem, an interacting measure, floor or regulator removal, and infinite volume are not proved here. All statements use the fixed positive production floor, common real-even regulators, the strict whole-shell filtration, regular one-shot controls, exact non-aliased products, the R-066 finite-low convention, and smooth inhomogeneous Littlewood–Paley blocks for analysis. The physical sharp-cube regulator is not changed.

2 Canonical complete-packet Doob decomposition

Fix a terminal shell J . Let (g_r, d_r) be the value and derivative Gaussian increments and let

$$X_r = \sum_{\ell \leq r} g_\ell, \quad DX_r = \sum_{\ell \leq r} d_\ell. \quad (2.1)$$

Write the partial control shifts, including the fixed low block, as

$$A^{(k)} = A_{<j_0} + \sum_{j_0 \leq \ell \leq k} a_\ell, \quad C^{(k)} = DA^{(k)}, \quad (2.2)$$

and set $A^{(j_0-1)} = A_{<j_0}$. Since a_ℓ is $\mathcal{F}_{\ell-1}$ -measurable, $A^{(k)}$ and $C^{(k)}$ are $\mathcal{F}_{k-1}$ -measurable.

For any shift H measurable before the next Gaussian shell define

$$\mathcal{W}_r^{(J)}(H) := W_r^{(J)}(X_r + H, DX_r + DH), \quad (2.3)$$

where $W_r^{(J)}$ is the R-066 terminal-backward heat energy. The terminal increment created by control shell k is

$$\Phi_k = \mathcal{W}_J^{(J)}(A^{(k)}) - \mathcal{W}_J^{(J)}(A^{(k-1)}). \quad (2.4)$$

2.1 Exact theorem

For $j_0 \leq k \leq r \leq J$ put

$$\begin{aligned} \mathfrak{G}_{r,k} := & \{ \mathcal{W}_r^{(J)}(A^{(k)}) - \mathcal{W}_r^{(J)}(A^{(k-1)}) \} \\ & - \{ \mathcal{W}_{r-1}^{(J)}(A^{(k)}) - \mathcal{W}_{r-1}^{(J)}(A^{(k-1)}) \}. \end{aligned} \quad (2.5)$$

Iterating the R-066 backward-heat martingale from J down to r gives

$$\mathbb{E}[\Phi_k \mid \mathcal{F}_r] = \mathcal{W}_r^{(J)}(A^{(k)}) - \mathcal{W}_r^{(J)}(A^{(k-1)}). \quad (2.6)$$

Consequently

$$\mathfrak{G}_{r,k} = \mathbb{E}[\Phi_k \mid \mathcal{F}_r] - \mathbb{E}[\Phi_k \mid \mathcal{F}_{r-1}], \quad (2.7)$$

and therefore

$$\boxed{\mathbb{E}[\mathfrak{G}_{r,k} \mid \mathcal{F}_{r-1}] = 0.} \quad (2.8)$$

The finite Doob sum is the exact identity

$$\boxed{\Phi_k = \mathfrak{P}_k + \sum_{r=k}^J \mathfrak{G}_{r,k}, \quad \mathfrak{P}_k := \mathcal{W}_{k-1}^{(J)}(A^{(k)}) - \mathcal{W}_{k-1}^{(J)}(A^{(k-1)})}. \quad (2.9)}$$

In particular,

$$\boxed{\mathbb{E} \sum_{k=j_0}^J \sum_{r=k}^J \mathfrak{G}_{r,k} = 0} \quad \text{for every } J. \quad (2.10)$$

No dyadic estimate or probability moment is used in (2.10). Summing the predictable baselines $\mathfrak{P}_k$ gives exactly the R-066 controlled discrete Ito formula. Thus the fresh-root branch is not a new absolute Carleson estimate: in the complete signed endpoint it is already the martingale part removed by the backward-heat coordinate.

This statement is deliberately weaker than

$$\sup_J \mathbb{E} \sum_{k \leq r} |\mathfrak{G}_{r,k}| < \infty. \quad (2.11)$$

Equation (2.11) is neither needed nor proved. Taking absolute values before the conditional cancellation would return to the critical architectures excluded by R-073 and R-076.

3 Why the complete Wick forest must stay inside the packet

At finite cutoff, condition on $\mathcal{F}_{r-1}$. Both partial shifts in (2.5) are now deterministic relative to the fresh shell r . The heat-averaged coefficient increment has the form

$$F_{r,k}^{(J)}(x) = B_r^{(J)}(x + A^{(k)}) - B_r^{(J)}(x + A^{(k-1)}). \quad (3.1)$$

The R-063 Taylor/Wick algebra applies conditionally. With its root-shell notation, the complete coefficient bracket is

$$\begin{aligned} \sum_{\ell} \{ & F_{r,k}^{(J)}(x_{\ell}) : Q_{\ell} \\ & + DF_{r,k}^{(J)}(x_{\ell}) : (P_3 + P_1) \\ & + \frac{1}{2} D^2 F_{r,k}^{(J)}(x_{\ell}) : (P_4 + P_2^{\text{cross}} + \Sigma_{\ell} Q_{\ell} + P_0^{\text{cross}}) \\ & + \mathcal{R}_{3,\ell}^{F_{r,k}^{(J)}} : Q_{\ell} \}. \end{aligned} \quad (3.2)$$

Here (3.2) is a finite-cutoff conditional equality, not a claim that an arbitrary adapted terminal coefficient has a finite global chaos expansion. It is accompanied by the derivative-current pieces of $\mathcal{W}$.

The scalar identities expose the issue:

$$Z(Y^2 - \gamma) =: ZYY : + 2kY, \quad (3.3)$$

$$Z^2(Y^2 - \gamma) =: ZZYY : + \sigma(Y^2 - \gamma) + 4k : ZY : + 2k^2. \quad (3.4)$$

The lower terms in (3.3)–(3.4) are generally nonzero. A contraction may turn a nominal top Gaussian pair into ΣQ , a second chaos, or a zero chaos. It is therefore false to give the top chaos one causal owner and discard or separately center its companions.

R-073 restores both first variations and the terminal square before raw-to-Wick conversion. R-066 transports the covariance trace and retains the finite-low boundary. Equations (2.5)–(2.10) apply to that complete packet. They do not assert

$$\mathbb{E}P_1 = \mathbb{E}P_2 = \mathbb{E}(\Sigma Q) = \mathbb{E}P_0 = 0 \quad (3.5)$$

term by term. This is the precise sense in which the R-063 forest “reassembles after conditioning.”

Combining (2.10) with the already proved R-066 trace and low-block bounds gives, for every $\eta, \zeta > 0$ and the corresponding fixed j_0 ,

$$\mathbb{E}\{\mathcal{F}_J^{\text{fresh}} + \mathcal{T}_J + \mathcal{L}_{<j_0}\} \geq -\eta \mathbb{E} \sum_j \|h_j\|_{\mathcal{H}_j}^2 - \zeta \mathbb{E} \|Z_J\|_6^6 - C_{\eta,\zeta,j_0}, \quad (3.6)$$

uniformly in J , where $\mathbb{E}\mathcal{F}_J^{\text{fresh}} = 0$. The signed trace and finite-low estimates, not a fictional termwise forest centering, supply the right side.

4 Payload-comparable resonance theorem

Use a smooth inhomogeneous Littlewood–Paley analysis partition $(\Delta_m)_{m \geq -1}$. This is an analysis device; the physical regulator may remain the pinned sharp cube. Let

$$J = M(U)^T DU, \quad K_t = D^3 M(U + tA) - D^3 M(U). \quad (4.1)$$

At the fixed positive floor,

$$\sup_{t \in [0,1]} \|K_t\|_{\infty} \leq 2 \sup_z \|D^3 M(z)\| < \infty. \quad (4.2)$$

R-050/R-075 give, for $s = 1/2 + \kappa < 1$ and every finite p ,

$$R := \|J\|_{\mathcal{C}^{-s}} \in L^p(\Omega). \quad (4.3)$$

Let the R-076 input-maximum expansion, with a fixed labelled tie rule, be

$$A^3 DA = \sum_{r \geq -1} G_r. \quad (4.4)$$

Every G_r contains each four-input monomial exactly once and has Fourier support in a ball of radius $C2^r$, even when its inputs make a low output. For a fixed resonance width L , define directly at finite cutoff

$$T_{\leq} := \frac{1}{2} \int_0^1 (1-t)^2 \sum_r \sum_{\substack{|m-n| \leq L \\ m \leq r+L}} \int \langle (\Delta_m K_t)(\Delta_n J), Q_{\Pi} G_r \rangle dx dt. \quad (4.5)$$

Formula (4.5), rather than a prior claim that $K_t \circ J$ exists, is the definition of the cutoff trilinear form.

4.1 Block estimate

Smooth analysis blocks are uniformly bounded on L^∞ , and (4.3) gives

$$\|\Delta_m K_t\|_\infty \lesssim_e 1, \quad \|\Delta_n J\|_\infty \lesssim 2^{sn} R. \quad (4.6)$$

Since $s > 0$,

$$\sum_{m \leq r+L} \sum_{|m-n| \leq L} 2^{sn} \leq C_{s,L} 2^{sr}. \quad (4.7)$$

Thus

$$\boxed{|T_{\leq}| \leq C_{e,s,L} R \sum_r 2^{sr} \|G_r\|_1.} \quad (4.8)$$

The estimate is pathwise. It uses no independence, Malliavin derivative, or control-dependent multiplier norm. Adaptation is harmless in this orientation because only the uniform bound (4.2) is used.

R-076 already proved

$$\sum_r 2^{sr} \|G_r\|_1 \lesssim \|A\|_{H^2}^{(1+s)/2} \|A\|_6^{(7-s)/2}. \quad (4.9)$$

Put $X = \|A\|_{H^2}^2$ and $Y = \|A\|_6^6$. Then

$$|T_{\leq}| \lesssim R X^a Y^b, \quad a = \frac{1+s}{4}, \quad b = \frac{7-s}{12}, \quad c = 1 - a - b = \frac{1-s}{6}. \quad (4.10)$$

One weighted Young inequality, after both shell sums, yields

$$\boxed{|T_{\leq}| \leq \eta X + \zeta Y + C_{e,s,L} \eta^{-\frac{3(1+s)}{2(1-s)}} \zeta^{-\frac{7-s}{2(1-s)}} R^{6/(1-s)}.} \quad (4.11)$$

At $\kappa = 1/10$ and $s = 3/5$,

$$(a, b, c) = \left(\frac{2}{5}, \frac{8}{15}, \frac{1}{15} \right), \quad \boxed{|T_{\leq}| \leq \eta X + \zeta Y + C_e \eta^{-6} \zeta^{-8} R^{15}.} \quad (4.12)$$

The fifteenth moment is already supplied by (4.3). No shellwise Young inequality is used, so the Cameron–Martin energy and sextic budget are each spent once.

The branch $m \leq r + L$ includes a unique maximal payload input, payload ties, mixed coefficient/payload ties, and the old pair-high cases. A second explicit high input is not required. High-high-to-low interactions inside G_r are retained because (4.8) uses the input scale and L^1 norm, not an output projection.

4.2 A valid but dominated pair-high refinement

One explored route first extracted a restricted subfamily with two comparable explicit high inputs. If a fraction θ of the companion shell's Cameron–Martin gain is used before sequence Holder, it gives

$$|T_{\text{pair}}| \lesssim N_{j_0}^{-\theta} R X^\alpha Y^\beta, \quad \alpha = \frac{1+s+2\theta}{4}, \quad \beta = \frac{7-s-2\theta}{12}. \quad (4.13)$$

The admissible interval is $0 < \theta < (1-s)/2$. Choosing $\theta = (1-s)/4$ gives, at $s = 3/5$,

$$\theta = \frac{1}{10}, \quad \alpha = \frac{9}{20}, \quad \beta = \frac{31}{60}, \quad 1 - \alpha - \beta = \frac{1}{30}, \quad (4.14)$$

and hence

$$|T_{\text{pair}}| \leq \eta X + \zeta Y + C \eta^{-27/2} \zeta^{-31/2} N_{j_0}^{-3} R^{30}. \quad (4.15)$$

This is a valid restricted refinement and records the earlier no-reuse search. It is not the primary proof: (4.12) covers the entire payload-comparable block with the smaller fifteenth moment and without extracting a pair. The refinement may still be useful if an explicit high-floor decay is later needed.

5 The exact coefficient-dominant residual

The complement of (4.5) is

$$T_{>} := \frac{1}{2} \int_0^1 (1-t)^2 \sum_r \sum_{\substack{|m-n| \leq L \\ m > r+L}} \int \langle (\Delta_m K_t)(\Delta_n J), Q_{\Pi} G_r \rangle dx dt. \quad (5.1)$$

Equations (4.5) and (5.1) are a disjoint exhaustive partition of the shifted resonance at every finite cutoff. Equality is assigned to $T_{\leq}$.

Because G_r is ball-supported, a fixed enlargement L_* gives

$$T_{>} = \sum_r \int S_{r+L_*} \left[\sum_{\substack{|m-n| \leq L \\ m > r+L}} (\Delta_m K_t)(\Delta_n J) \right] G_r, \quad (5.2)$$

with the path weight and tensor contractions suppressed. This is a genuine high–high-to-low form. Its two high factors may create arbitrarily low output. The elementary carrier

$$e^{iNx} e^{-iNx} = 1 \quad (5.3)$$

shows that no deterministic separation decay follows from $m - r \rightarrow \infty$. The zero-mode fixture in R-076 is the corresponding production-frame diagnostic.

The proper remaining object is not isolated $T_{>}$. Let $\mathfrak{B}_J^>$ denote the nonduplicating predictable block containing

1. $T_{>}$ and the base-frozen term not already paid by R-075/R-076;
2. both R-073 restored first variations;
3. the corresponding terminal-square polarization;
4. coefficient curvature and the complete R-063 P_3/P_1 and $P_4/P_2/\Sigma Q/P_0$ conversion;
5. R-066 trace transport and the finite-low boundary.

The Doob projection of Section 2 may be applied only to complete fresh-root packets inside this list. Its conditional compensator, Wick contractions, mixed ties, and coefficient-control-high pieces remain in $\mathfrak{B}_J^>$. The smallest honest successor is

$$\boxed{\mathbb{E}\mathfrak{B}_J^> \geq -\eta\mathbb{E}\sum_j \|h_j\|_{\mathcal{H}_j}^2 - \zeta\mathbb{E}\|Z_J\|_6^6 - C_{\eta,\zeta,e,j_0}} \quad (5.4)$$

uniformly in J , for regular one-shot strict-past controls. If (5.4) is proved, R-075 graph recovery supplies the fixed-cutoff finite-energy extension and R-066 completes controlled-shell one-use. Only then may the existing $q = 10/9$ equivalence be used for Nelson synthesis.

6 Correction of the old root map and failed routes

The exact proof map after this note is sequential, not a raw three-coloring of monomials:

1. Decompose every complete terminal shell increment by (2.9). The Doob part has zero signed expectation; retain the predictable baseline.
2. In the exact endpoint representation of that baseline, split only the shifted resonance by (4.5) and (5.1). Pay $T_{\leq}$ once by (4.11).
3. Keep $T_{>}$ with every signed companion listed below (5.3), and prove the single lower bound (5.4).

This ordering prevents duplication: Section 2 acts on the complete endpoint packet, while Section 4 subtracts one proved form from its predictable baseline.

The investigated alternatives are now completely recorded.

1. **Raw monomial largest-root ownership fails as a theorem.** Coefficient/payload ties have no declared unique owner, and Wick contraction can remove the nominal largest Gaussian pair. This is recorded as AUDIT-2026-07-25-A13-R076-ROOT-TAXONOMY. The proposal in R-076 was explicitly open; the correction changes no earlier proved identity.
2. **Termwise forest centering fails.** Equations (3.3)–(3.4) contain nonzero lower chaoses. Only the complete packet has the martingale cancellation (2.10).
3. **Absolute fresh-root summability is unnecessary and unproved.** The required variational quantity is signed expectation. Replacing (2.10) by (2.11) discards the load-bearing cancellation.
4. **A standalone shifted resonance is not constructed.** $L^\infty \times \mathcal{C}^{-s}$ does not canonically define $K_t \circ J$. Equation (4.5) defines and bounds the cutoff trilinear form directly.
5. **Raw sharp projectors are not used in (4.6).** Their L^∞ operator norms are not uniformly controlled. Smooth analysis blocks supply the required uniform kernel bound without altering the physical sharp regulator.
6. **Output support does not kill $T_{>}$.** Equation (5.3) is an exact counterexample. The low-output insertion in (5.2) isolates rather than removes the obstruction.
7. **The pair-high harvest is valid but dominated.** It gives (4.15), but the simpler geometric sum proves (4.12) for the whole payload-comparable orientation with a smaller moment.
8. **Gaussian integration by parts is not a substitute.** Applied to a later adapted control it differentiates an uncontrolled Malliavin direction. The Doob packet uses only strict-past measurability and avoids that derivative.

7 Devil’s-advocate review

1. **“Each R-063 chaos centers separately.” UPHELD as false.** The lower-chaos polynomials in (3.3)–(3.4) are nonzero. Equation (2.8) is asserted only for the complete endpoint packet.
2. **“Doob cancellation proves an absolute Carleson estimate.” UPHELD as false.** It proves (2.10), not (2.11). The signed expectation is exactly the quantity needed at this stage.
3. **“Later adapted controls can be frozen before every earlier root.” UPHELD as false.** Partial controls are frozen only after the causal shell telescope. Formula (2.9) never freezes the terminal shift wholesale.
4. **“The $T_{\leq}$ estimate assumes probabilistic independence from the control.” DISMISSED.** It is pathwise. Control independence means only that $R = \|J\|_{C^{-s}}$ has a cutoff-uniform moment not growing with the chosen control.
5. **“A bounded K_t defines $K_t \circ J$.” UPHELD as false.** Only the oriented finite trilinear form (4.5) is defined. This avoids repeating the separated-multiplier error of R-076.
6. **“High–high-to-low inside the payload was discarded.” DISMISSED.** The input-maximum G_r blocks retain it. No output projection appears in (4.8)–(4.9).
7. **“High–high-to-low in the multiplier/current product is also harmless.” UPHELD as false.** It is exactly $T_{>}$, and (5.3) shows why spatial separation gives no decay.
8. **“Every payload tie contains two maximal explicit factors.” UPHELD as false.** A unique maximal input and mixed ties occur. The proof of (4.8) needs no pair extraction.
9. **“The optional thirtieth-moment refinement is the main closure.” DISMISSED.** The broader fifteenth-moment theorem (4.12) is stronger for the present gate. The restricted refinement is kept only as an audited route and possible future tail tool.
10. **“Smooth analysis blocks change the sharp-cube model.” DISMISSED.** They are used only for Bony analysis of an already regularized product. The physical covariance and filtration remain the pinned sharp cubes.
11. **“This closes controlled-shell one-use or Sector A.” UPHELD as an overclaim.** The complete coefficient-dominant bound (5.4), one-use, and Nelson synthesis remain open. The tier stays T4.

8 Reproduction and independent audit

The primary executable derives every exponent from $\kappa = 1/10$, proves an exact two-shell Gaussian backward-heat/Doob polynomial fixture, checks the complete scalar Wick forests, exhaustively partitions finite block-index triples, evaluates the high–high zero mode, and tests the weighted AM–GM normalization. It also derives the optional pair-high exponents rather than copying them.

The non-importing executable uses a finite probability tree for Doob centering, correlated-normal arrays for the forest identities, a separately implemented block enumeration, an independent exponent route, and complex opposite-frequency carriers. It does not import the primary executable.

Run from the repository root:

```
python codes/foundations/a13_classii_causal_packet_payload_resonance_verify.py
```

The integrated verifier is fail-closed: it pins the R-063, R-066, R-069, R-073, R-075, and R-076 authorities; reruns both executables; checks the proof source, PDF, manifest, and repository surfaces; and rejects any overclaim of (5.4), one-use, Nelson, or Sector-A closure.

Result footer

Result ID: A13-CLASSII-CAUSAL-PACKET-PAYLOAD-RESONANCE-REDUCTION

Precise statement: Complete fresh-Gaussian Doob packets have exactly zero signed expectation. The complete smooth-LP payload-comparable shifted-resonance block obeys the R-076 one-use Besov payment and, at $s=3/5$, requires only the available fifteenth moment. The complement is one exact coefficient-dominant high-high-to-low signed packet.

Scope: Fixed positive floor; finite cutoff and uniform cutoff estimates; common real-even regulators; strict whole-shell one-shot controls; smooth LP analysis blocks; R-066 finite-low convention.

Dependencies: A1; R-050; R-063; R-066; R-069; R-073; R-075; R-076.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: `python codes/foundations/a13_classii_causal_packet_payload_resonance_verify.py`

Expected output: primary 35/35 + independent 26/26 + manifest-pinned integrated/aggregate PASS.

Falsification gate: Nonzero conditional mean in (2.8); missing lower chaos in (3.2); failure of (4.7); disagreement of $2/5$, $8/15$, $1/15$, or moment 15; overlap/gap in the two-way split; disappearance of (5.3); or any authority, source, PDF, run, or surface mismatch.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: The coefficient-dominant inequality (5.4), every-finite-energy extension, controlled-shell one-use, $q=10/9$ Nelson, interacting measure, all removal limits, and Sector-A closure remain open.

Next required action: Prove the complete coefficient-dominant $m>r+L$ high-high-to-low signed packet with both first variations, terminal square, curvature, full Wick forest, trace, and finite-low boundary retained.

Evidence summary: Exact Doob/tower proof; complete conditional Wick forest; analytic trilinear/Besov/Young proof; primary and non-importing executable audits; PDF.